



YISHUN INNOVA JUNIOR COLLEGE
JC 1 PROMOTIONAL EXAMINATION
Higher 2

CANDIDATE
NAME

CG

INDEX NO

COMPUTING

Paper 1 Written

9569/01

28 Sep 2023

1 hour 10 minutes

Additional Materials: Writing Paper
 Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class clearly on the cover page.
Write in dark blue or black pen on the writing paper provided.
You may use an HB pencil for any diagrams, graphs, tables or rough working.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.
Approved calculators are allowed.

At the end of the examination, fasten your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is **40**.

- 1** All network devices have a unique Media Access Control (MAC) address encoded into its hardware. It consists of 12 hexadecimal digits.

An example is shown below:

1F:A3:98:9A:FE:4E

The first 3 bytes of the MAC address is known as the Organisationally Unique Identifier (OUI), which is used to identify the device's manufacturer. The remaining bytes of the MAC address are unique and they are used to identify each device.

- (a) State the total number of bytes used for a MAC address. [1]
- (b) Convert the OUI portion of the given MAC address into a binary number. [2]
- (c) Calculate the maximum number of unique MAC addresses which can be generated for each OUI. [2]

- 2** A library wants to arrange of books according to their unique 3 digit ID numbers. An example of seven ID numbers are given below:

ID = [542, 304, 215, 391, 482, 731, 629]

- (a) Describe the Bubble Sort algorithm. [2]
- (b) Explain how the Bubble Sort algorithm can be optimised. [1]

The other sorting algorithm being considered is the Quicksort algorithm.

(c) State the ideal pivot for the Quicksort algorithm to execute most efficiently. [1]

The following recursive non in-place Quicksort function is used to sort the elements in a list `lst` in ascending order. The function is currently incomplete and has four missing sections (A, B, C, and D).

The first element of the list `lst` is being selected as the pivot:

```
01 FUNCTION QUICKSORT(lst) RETURNS LIST
02 IF LENGTH(lst) <= 1 THEN
03     RETURN lst
04
05 PivotIndex =     A    
06 PivotValue =     B    
07 Left = []
08 Middle = []
09 Right = []
10
11 FOR     C    :
12     IF element < PivotValue THEN
13         APPEND element TO Left
14     ELSE IF element > PivotValue THEN
15         APPEND element TO Right
16     ELSE
17         APPEND element TO Middle
18
19 // Recursively call the Quicksort function
20 RETURN     D    
```

(d) Write the missing pseudocode for **A**, **B**, **C** and **D** in the function. [3]

ID = [542, 391, 215, 482, 304, 731, 629]

- (e) State the index of the pivot and the arrangement of the ID numbers in the above list after partitioning the list once. [2]
- (f) State **one** advantage and **one** disadvantage of using this non in-place Quicksort algorithm instead of the Bubble Sort algorithm to arrange a large collection of the ID numbers. [2]

- 3 In a newly established company, the employees work using standalone personal computers and share their files using removeable storage drives.

The company is considering to set up a local area network (LAN) with a file server for the employees.

- (a) Describe **two** benefits of storing files in a file server within a LAN . [2]
- (b) Besides file sharing, suggest **two** additional functions that can be provided within the LAN. [2]

The company is also considering to install a router and a modem to provide internet access via an Internet Service Provider (ISP). This will also allow the employees to access the LAN remotely.

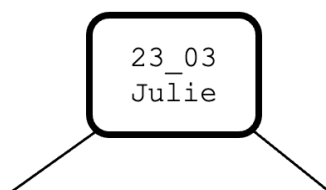
- (c) Describe **one** advantage and **one** disadvantage of providing remote access for the employees. [2]
- (d) Describe the functions of the router and the modem. [2]

- 4 The table shows the names and matriculation numbers of 8 students:

Matriculation Number	Name
23_03	Julie
23_07	Samantha
23_01	Xavier
23_04	Beatrice
23_02	Peter
23_05	Alfred
23_06	Teresa
23_08	John

A binary search tree (BST) with the matriculation number as the key will be used to store the students' data.

The following incomplete BST diagram shows the data of the first student being inserted into the root node:



- (a) The remaining students' data are to be inserted into the BST according to the order given in the above table. Copy and complete the BST diagram. [3]
- (b) Describe the algorithm to perform a search for a particular student's data in the BST. [3]
- (c) Compare the time complexities for performing a search for a student's data in a BST and an unsorted list. [2]
- (d) Describe the traversal in the BST to obtain a list of students' data arranged in an ascending order of the matriculation numbers. [2]

- 5** A vehicle workshop has numerous customers who utilise its repair and maintenance services.

The details of each vehicle are written on a record card. All the record cards are kept in a filing cabinet and sorted in ascending order of the vehicles' registration numbers.

It is difficult and time consuming to linearly search through the record cards in the filing cabinet.

- (a)** State and describe a method of retrieving the record card of a particular vehicle from the filing cabinet. [2]

To improve the filing system for easier retrieval and addition of vehicle record cards, a hashing algorithm is to be used to store and retrieve a record card.

- The filing cabinet has one hundred files labelled from 1 to 100.
 - A hashing algorithm will use the vehicle's registration number to calculate a hash value, ranging between 1 and 100.
 - The record cards will be kept in separate files according to the hash values calculated.
- (b)** Describe how the record cards for different vehicles with the same hash value can be managed. [2]
- (c)** Explain the advantage of using a hashing algorithm in this improved filing system. [2]

END OF PAPER